

The Trainer v0.1 - Freeze and Forensic Dossier

Document status: functional baseline final amendment

Final amendment: 2026-07-04

Audit branch: audit/trainer-v0.1-freeze

Read-only source branch: headless-gsplat-v0.1-dev

1. Executive summary

The final Trainer v0.1 baseline is runtime commit 221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3, tree f26a1a2fddd2f0a3398b3b49b5fbb61ea24038d5, and Docker digest sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98.

Its real end-to-end execution was confirmed by the repository owner and operator:

```
validation_status: passed
validation_authority: repository-owner-operator
evidence_level: operator-confirmed
primary_execution_records: not-preserved
```

The confirmed final flow includes Surveyor archive transfer, checksum verification, automatic transactional preparar -escena, dataset check, CUDA training, checkpoint, PLY generation, packaging, result transfer, and viewer load. Primary logs, transcript, checkpoint, packages, and PLY were not preserved, so this confirmation is not classified as primary evidence and carries no inferred metrics.

The original Prueba02 run remains separate primary evidence for an earlier runtime:

- source commit: 90196251d59907754cc19caf76b59a737752893b;
- source tree: 9568224839eb12bfd0bfbca4625bdea3bfe7a8bb;
- Docker digest associated by workflow chronology: sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89;
- evidence: manual extraction, 300-step training, ckpt_299_rank0.pt, PLY inspection, packaging, transfer, and SuperSplat v2.27.4.

Prueba02 metrics are not inherited by the final run. The older digest remains a historical artifact and is not the stable Trainer v0.1 digest.

Proposed immutable identifiers:

- Git tag trainer-v0.1.0-smoke-validated targeting commit 221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3;
- Docker tag headless-gsplat-v0.1.0 targeting digest sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98 without rebuilding.

No Git tag or Docker tag was created by this audit.

2. Evidence language and confidence

This dossier uses four evidence levels:

- **Validated:** supported by versioned evidence, a preserved workflow log, an executable local test, or an inspected artifact.
- **Reported:** preserved as an operator observation, but the original raw artifact or log is unavailable.
- **Inferred:** derived from immutable chronology or source behavior and explicitly identified as inference.
- **Operator-confirmed:** pass/fail outcome confirmed by the repository owner and operator without preserved primary execution records.

This distinction is important because the repository preserves the Prueba02 report and screenshot, but not the large scene archive, checkpoint, PLY files, full pod transcript, or pod-side image digest.

3. Git identity and chain of custody

Item	Value
Remote fetch/push	https://github.com/Marquezmemo/cloud-workstation.git
Source branch	headless-gsplat-v0.1-dev
Source HEAD recorded before audit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Source tree	f26a1a2fddd2f0a3398b3b49b5fbb61ea24038d5
Audit branch	audit/trainer-v0.1-freeze
Audit branch base	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Final documentation commit	123652ea97f549e1d06d3993f55c2bcab252f840
Final baseline runtime commit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Final baseline runtime tree	f26a1a2fddd2f0a3398b3b49b5fbb61ea24038d5
Historical Prueba02 runtime	90196251d59907754cc19caf76b59a737752893b

The tracked source tree had no diff at audit start. The original shared worktree contained an unrelated untracked output/ directory with Surveyor freeze artifacts. It was not modified, moved, staged, or copied. The audit was created in an isolated worktree based on the exact source HEAD.

The audit changes only this dossier and its generated artifacts. Functional paths such as Dockerfile, workflows, requirements, scripts, tests, trainer, patch, exporter, and packaging remain byte-for-byte equal to source HEAD 221c4ef.

4. Docker identity

4.1 Final v0.1 image

Item	Value
Workflow	Publish Headless gsplat Dev
Workflow run	https://github.com/Marquezmemo/cloud-workstation/actions/runs/28413758338
Result	success
Source commit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Tag configured by workflow	docker.io/\${DOCKERHUB_USERNAME}/cloud-workstation:headless-gsplat-v0.1-dev
Manifest digest	sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98
Platform	linux/amd64
Validation	operator-confirmed; primary records not preserved

4.2 Historical Prueba02 publication

Item	Value
Workflow run	https://github.com/Marquezmemo/cloud-workstation/actions/runs/28285526569
Source commit	90196251d59907754cc19caf76b59a737752893b
Manifest digest	sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89
Relationship to final baseline	historical runtime with preserved Prueba02 evidence

The failed documentation-triggered workflow run 28411332932 did not replace the development tag. The next successful run published the final 221c4ef image.

4.3 Base image

```
runpod/pytorch:2.4.0-py3.11-cuda12.4.1-devel-ubuntu22.04
@sha256:61a4aafb0094cd773f11eefa378929d5a687bd775febeb78eac62fc824141fb5
```

The base image is pinned by manifest digest. The Trainer does not use latest.

5. Runtime and fixed dependencies

5.1 Runtime identity

Component	Identity	Evidence note
Ubuntu	22.04 base	Dockerfile/base identity
Python	3.11 line; 3.11.10 previously observed	Earlier RunPod runtime evidence; not separately recaptured in the Prueba02 record
PyTorch	base 2.4.0 line; 2.4.1+cu124 previously observed	Earlier RunPod runtime evidence
CUDA toolkit/runtime in Trainer	12.4 line	Base tag and earlier runtime evidence
GPU used by Prueba02	NVIDIA GeForce RTX 4090	Operator and validation record
gsplat	1.5.3	Exact requirement and build label
OpenCV	opencv-python-headless==4.10.0.84	Exact requirement
runpodctl	2.5.0	Docker build ARG and verified binary checksum

The Surveyor runtime had a different CUDA stack. Its reported CUDA versions do not redefine the Trainer image identity.

5.2 Core Python requirements

```
gsplat==1.5.3
jaxtyping==0.3.11
ninja==1.13.0
numpy==1.26.3
rich==14.3.4
typing_extensions==4.15.0
wadler-lindig==0.1.7
```

5.3 Official trainer requirements

```
fused-ssim @ git+https://github.com/rahul-goel/fused-ssim@328dc9836f513d00c4b5bc38fe30478b4435cbb5
imageio[ffmpeg]==2.37.3
matplotlib==3.11.0
numpy==1.26.3
opencv-python-headless==4.10.0.84
piexif==1.1.3
pycolmap @ git+https://github.com/rmbruailla/pycolmap@cc7ea4b7301720ac29287dbe450952511b32125e
pyyaml==6.0.2
scikit-learn==1.4.2
scipy==1.12.0
tensorboard==2.20.0
torchmetrics[image]==1.9.0
```

```
tqdm==4.68.2
tyro==1.0.13
```

Direct dependencies are pinned, including both Git dependencies by commit. Transitive dependencies are not fully locked with hashes. Historical `pip-freeze.txt` and `dpkg-freeze.txt` belong to the earlier workstation baseline and are not treated as a complete freeze of the Prueba02 Trainer image.

5.4 System packages added by Trainer

```
git cmake ninja-build build-essential ffmpeg wget curl unzip nano httpd tmux ca-certificates
```

6. Official source and exact headless patch

The training backend is the official gsplat example:

```
repository: https://github.com/nerfstudio-project/gsplat
release line: v1.5.3
commit: 937e29912570c372bed6747a5c9bf85fed877bae
path: examples/simple_trainer.py
installed path: /opt/gsplat/examples/simple_trainer.py
```

The build removes the cloned `.git` directory after checking out the fixed commit.

`patch-gsplat-simple-trainer-headless.sh` performs only these source substitutions:

1. Replaces `import viser` with `viser = None` and a headless comment.
2. Replaces `GsplatViewer` and `GsplatRenderTabState` imports with local stubs. Constructing the viewer stub raises an explicit headless error.
3. Replaces `CameraState`, `RenderTabState`, and `apply_float_colormap` imports with object/function stubs. Calling the colormap stub raises an explicit headless error.
4. Verifies that the three top-level viewer import forms no longer remain.

The patch does not alter training math, camera parsing, optimizer behavior, checkpoint serialization, PLY export, or dataset loading. The wrapper also passes `--disable_viewer` and rejects `DISABLE_VIEWER=false`.

7. Training architecture

```
Surveyor package
-> preparar-escena <scene>-surveyor-scene.tar.gz
-> transactional /workspace/scenes/<scene>
-> train-scene.sh --check <scene> (invoked by importer)
-> official simple_trainer.py default --disable_viewer
-> CUDA training
-> /workspace/outputs/<scene>/ckpts/ckpt_<step>_rank<rank>.pt
-> generar / generate-ply.py
-> standard and compressed PLY
-> empaquetar
-> runpodctl transfer
-> independent inspection / SuperSplat
```

The official trainer initializes CUDA device `cuda:<local_rank>`, parses COLMAP data, creates splats and optimizers, trains for `max_steps`, and writes the final checkpoint at `max_steps - 1`. This explains `MAX_STEPS=300` producing `ckpt_299_rank0.pt`.

The checkpoint contains at least:

```
step
splats: means, scales, quats, opacities, sh0, shN
```

Optional pose or appearance state is added only when the corresponding features are enabled.

8. Workspace, command surface, and persistence

8.1 Final baseline layout at commit 221c4ef

```
/workspace/datasets
/workspace/scenes
/workspace/outputs
/workspace/logs
/workspace/checkpoints
/workspace/temp
```

Persistence is not provided by directory creation alone. Datasets, scenes, outputs, logs, checkpoints, and transfer artifacts are persistent only when `/workspace` is backed by RunPod persistent storage or a mounted volume.

8.3 CMD

```
CMD ["runpod-keepalive.sh"]
```

The `keepalive` command creates `/workspace/logs`, prints validation hints, and runs `sleep infinity` so a non-interactive pod remains available.

8.4 Commands

Command	Responsibility	Final v0.1 image
<code>prepare-dataset.sh</code>	Creates an empty COLMAP scene skeleton	present
<code>train-scene.sh --check</code>	Structural dataset gate and run metadata	present and exercised
<code>train-scene.sh</code>	Wraps official trainer, logs and checkpoint alias	present and exercised
<code>generar / Generar</code>	Calls <code>generate-ply.py</code>	present and exercised
<code>empaquetar</code>	Archives PLY exports and writes checksum	present and exercised
<code>comprimir</code>	Alias for <code>empaquetar</code>	present
<code>empaquetar -logs</code>	Packages scene-scoped existing evidence with portable checksum	present; not part of preserved Prueba02 sequence
<code>collect-training-diagnostics.sh</code>	Captures runtime, GPU, environment, trees and recent logs	present
<code>validate-gpu.sh</code>	Verifies NVIDIA, torch, CUDA operation and gsplat import	present
<code>preparar -escena</code>	Securely verifies and imports Surveyor archive	present; regression-tested; operator-confirmed

9. Surveyor-to-Trainer scene contract

Prueba02 arrived as `Prueba02-surveyor-scene.tar.gz` with an adjacent portable checksum record. The checksum was verified in Surveyor and after receipt by Trainer. The exact scene archive SHA-256 was not preserved in the attachments and is not reconstructed here.

Minimum Trainer scene structure:

```
/workspace/scenes/Prueba02/
  images/
  sparse/0/cameras.bin
  sparse/0/images.bin
  sparse/0/points3D.bin
  surveyor-manifest.json
```

Surveyor evidence records 30 `input/database/registered` images, one camera, `camera_model=OPENCV`, two sparse candidates, and selection of the 30-image, 4,555-point model normalized to `sparse/0`.

During Prueba02 original the archive was manually extracted before `train-scene.sh --check` passed. In the final baseline, `preparar -escena` verifies the portable checksum, inspects the tar and manifest, installs transactionally with rollback, and runs the productive dataset check. Its real use is operator-confirmed; primary execution records were not preserved.

10. Camera and real OPENCV undistortion behavior

Prueba02 used one COLMAP OPENCV camera. The fixed official parser does not ignore distortion.

For every camera it reads:

```
fx, fy, cx, cy
```

For OPENCV it also reads:

```
k1, k2, p1, p2
```

It constructs the initial intrinsic matrix K , then for perspective distortion:

1. Calls `cv2.getOptimalNewCameraMatrix(K, params, size, 0)`.
2. Calls `cv2.initUndistortRectifyMap(...)` with the corrected matrix.
3. Stores the corrected intrinsic matrix and valid ROI.
4. Updates the effective image size to the ROI dimensions.
5. In dataset loading, calls `cv2.remap(...)`.
6. Crops the remapped image to the valid ROI.
7. Supplies the corrected image and corrected K to training.

The warning `COLMAP Camera is not PINHOLE. Images have distortion.` reports the input model. It is not evidence that distortion is ignored.

Surveyor copies source images into the scene and runs feature extraction, matching, and mapping. It does not invoke COLMAP `image_undistorter`. Therefore v0.1 performs one undistortion in the fixed gsplat parser. It does not change the Surveyor camera to PINHOLE, and it does not perform a second undistortion.

Double-undistortion risk: a future implementation must change both halves of the contract together. If Surveyor starts delivering already-undistorted images but leaves OPENCV distortion parameters active, the Trainer parser will remap them again. If Surveyor changes the camera metadata without matching image geometry and corrected intrinsics, training will receive an inconsistent camera model.

Primary source:

```
https://github.com/nerfstudio-project/gsplat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/datasets/colmap.py
```

11. Checkpoints, outputs, and exports

`train-scene.sh` writes:

```
/workspace/logs/<scene>/run.env
/workspace/logs/<scene>/run.summary
/workspace/logs/<scene>/train.log
/workspace/logs/<scene>/error.tail      # failure only
/workspace/outputs/<scene>/...
/workspace/checkpoints/<scene>        # symlink to outputs/<scene>/ckpts
```

The official trainer creates:

```
ckpts/
stats/
renders/
ply/
tb/
```

`generate-ply.py` discovers or accepts an explicit checkpoint, loads tensors, normalizes required fields, uses `gsplat.export_splats`, and writes:

```
/workspace/outputs/<scene>/exports/<scene>.ply
/workspace/outputs/<scene>/exports/<scene>.compressed.ply
/workspace/logs/<scene>/generate-ply.log
```

The standard PLY gate requires binary little endian encoding and a vertex count equal to the checkpoint splat count. The compressed gate rejects a count larger than the input count.

`empaquetar` includes existing `.ply` files in `<scene>-ply-exports.tar.gz`. Its checksum record contains the absolute pod path in the validated v0.1 behavior. It can be checked in place; after transfer the digest must be compared manually. `empaquetar - logs` uses a relative, portable checksum but does not generate telemetry.

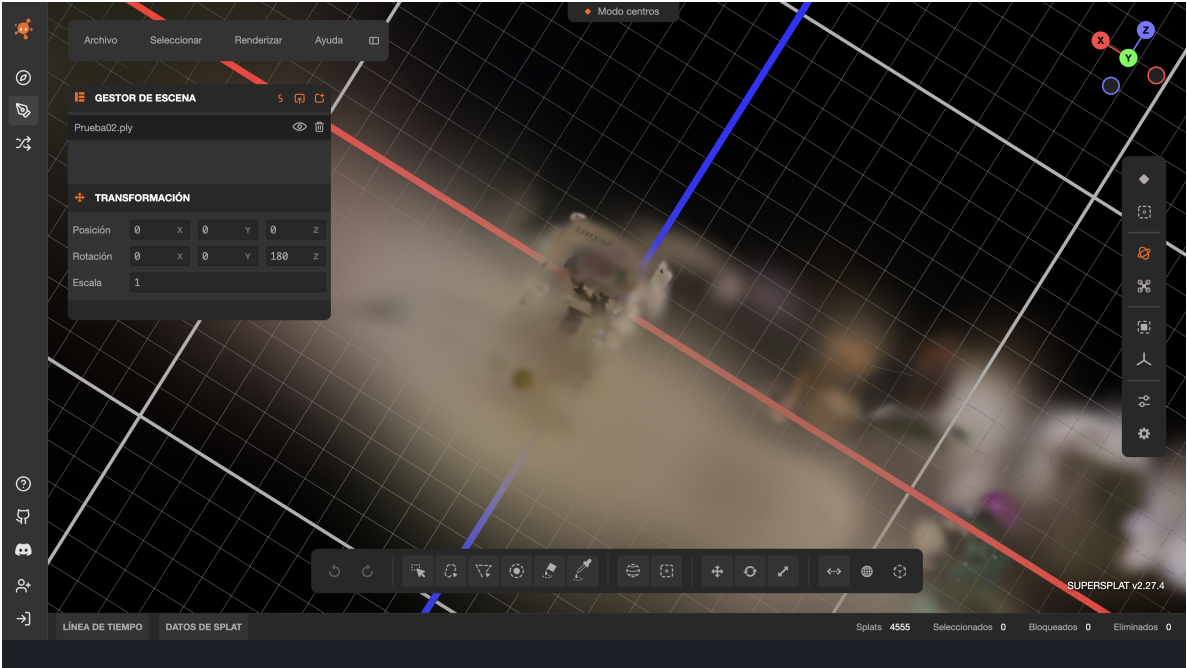
12. Prueba02 evidence

Observation	Result	Evidence level
Scene archive	Prueba02-surveyor-scene.tar.gz; checksum passed	Validated; exact archive digest not preserved
Scene archive size	741.1 MB; transfer UI showed 777 MB	Reported and versioned in Surveyor record
Images	30	Validated
Camera	1 x OPENCV	Validated
Training depth	MAX_STEPS=300	Validated
Duration	2:20	Validated record
Throughput	2.13 it/s	Validated record
Final reported loss	0.196	Reproducibility observation only; not a quality score
Initial Gaussians	4,555	Validated record
Final Gaussians	4,555	Validated record
Checkpoint	ckpt_299_rank0.pt	Validated record
Reported GPU memory	3.2356 GB	Validated record
Standard PLY	1,076,455 bytes	Validated record
Compressed PLY	280,894 bytes	Validated record
Standard vertices	4,555	Independently inspected
Compressed vertices	4,555	Export log/operator record
Standard non-finite values	0 NaN, 0 infinities	Independently inspected
Compressed non-finite values	0 NaN, 0 infinities	Operator assertion in freeze mission; raw compressed artifact/inspection output not versioned
SuperSplat	Standard PLY loaded and rendered as 4,555 splats in v2.27.4	Validated screenshot

The independently inspected standard PLY had SHA-256:

```
f7acbb332667a028756f7f2426dd07bfde577f66f30ef0dfa27b02449eea5c71
```

It was binary little endian, contained 59 float properties, had an exact payload for its header and vertex count, and contained no NaN or infinite values.



Prueba02 loaded in SuperSplat

The unchanged Gaussian count means the run did not validate densification. The screenshot proves compatibility and rendering, not professional geometry, commercial quality, capture quality, or production readiness.

12A. Final v0.1 Operator Validation

The repository owner and execution operator confirmed the final runtime flow for scene Prueba02:

```
validation_status: passed
validation_authority: repository-owner-operator
evidence_level: operator-confirmed
primary_execution_records: not-preserved
```

Field	Result
Runtime commit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Docker digest	sha256:726ac98c9678431fc34fbcc1de0bf07b71adbf215d672f04740a9599a6bf98
Transfer and checksum	passed
preparar -escena and transactional installation	passed
Dataset check and CUDA training	passed
Checkpoint and PLY	generated
Packaging, transfer and viewer load	passed
Primary logs, transcript and artifacts	not preserved
Professional/commercial quality	not evaluated
Densification	not certified

This attestation validates the stated outcomes only. It does not inherit Prueba02 metrics and does not authorize inference of the exact execution date, training depth, duration, performance, hashes, sizes, counts, or quality.

13. Tests executed for this freeze

No GPU training, model download, image build, or image publication was performed.

Non-destructive checks executed from the audit worktree:

```
bash -n for all existing Trainer shell entrypoints
bash tests/test-preparar-escena-trainer.sh
git diff --exit-code <source-head> -- Dockerfile workflows requirements scripts tests
```

Result:

```
All 15 preparar-escena tests passed.
Functional diff against source HEAD: zero.
```

The suite covers help/arguments, success through the productive `train-scene.sh --check`, missing files, checksum mismatch, path traversal, absolute paths, multiple roots, filename/root mismatch, invalid manifest, manifest consistency, COLMAP structure, existing destination, safe overwrite, rollback, capitalization, and source archive preservation.

These tests validate final runtime 221c4ef. Prueba02 original at 9019625 predates preparar-escena and remains a separate evidence-bearing execution.

14. Limitations

- Densification was not validated; the Gaussian count stayed at 4,555.
- Professional quality was not validated.
- Commercial quality was not validated.
- The loss value 0.196 is not a quality score.
- PSNR, SSIM, LPIPS, comparable renders, and a quality threshold were not preserved.
- The full scene archive, checkpoint, PLY files, compressed PLY inspector output, and full pod logs are outside Git.
- The exact scene archive digest was not preserved in the local attachments.
- The pod-side image digest was not captured. The validated digest association is based on immutable workflow chronology.
- Primary execution records for the final 221c4ef validation were not preserved.
- `empaquetar` writes an absolute-path checksum in the validated image.
- GPU telemetry was not continuously packaged by Trainer.
- Files above 1 GB, interrupted transfers, resume behavior, and measured transfer throughput remain unvalidated.
- The first run downloaded a 233 MB AlexNet model and spent 61.61 seconds setting up a CUDA extension; these are operational observations, not frozen performance guarantees.

15. Risks

1. **Moving development tag.** `headless-gsplat-v0.1-dev` is mutable; restoration must use the final digest.
2. **Evidence-level confusion.** The final operator confirmation must not be presented as preserved primary evidence.
3. **Unpinned transitive dependencies.** Rebuilding the same Dockerfile can produce a different manifest digest.
4. **Double undistortion.** Changing only Surveyor image geometry or only camera metadata can cause a second remap or inconsistent intrinsics.
5. **Evidence gaps.** Missing large artifacts prevent independent replay of every numerical claim.
6. **Short smoke depth.** A 300-step run proves pipeline function, not convergence or final quality.
7. **Diagnostics sensitivity.** Global diagnostics can contain environment data and must be handled as potentially sensitive evidence.

16. Recovery procedure

To recover the final v0.1 runtime without rebuilding:

1. Resolve the proposed Git tag target to commit 221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3.
2. Pull the image by digest, not by mutable development tag:

```
docker.io/${DOCKERHUB_USERNAME}/cloud-workstation@sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98
```

3. Mount persistent storage at /workspace.
4. Restore Prueba02-surveyor-scene.tar.gz and its adjacent checksum.
5. Run preparar-escena /workspace/Prueba02-surveyor-scene.tar.gz; it verifies the checksum and runs the dataset check.
6. Define an explicit MAX_STEPS for a new replay and preserve all primary records.
7. Export with generar --scene Prueba02, package with empaquetar Prueba02, and verify checksums at both ends.

To create the proposed stable Docker tag without rebuilding, an authorized registry operation should copy the existing digest to headless-gsplat-v0.1.0. The operation must verify the resulting tag resolves to the same manifest digest. This audit does not execute that mutation.

17. Handoff to The Eye

The Eye should integrate the following facts into official documentation without altering runtime behavior:

- freeze runtime source 221c4ef and digest sha256:726ac98...a6bf98 as the final baseline;
- retain 9019625 and sha256:535822...b63a89 only as historical Prueba02 identity;
- record the OPENCV remap/crop behavior and remove any implication that the warning means distortion is ignored;
- preserve the double-undistortion warning;
- state that preparar-escena is regression-tested and operator-confirmed in real operation, without preserved primary records;
- preserve Prueba02 numbers with evidence grades and limitations;
- do not claim densification, professional quality, or commercial quality;
- do not claim the compressed PLY non-finite scan is independently reproducible unless its raw inspection evidence is preserved;
- adopt the proposed Git and Docker tags only after verifying their exact targets;
- retain this report, JSON manifest, PDF, and SHA256SUMS as one evidence set.

18. Primary references

- Repository: <https://github.com/Marquezmemo/cloud-workstation>
- Historical Prueba02 image workflow: <https://github.com/Marquezmemo/cloud-workstation/actions/runs/28285526569>
- Final baseline image workflow: <https://github.com/Marquezmemo/cloud-workstation/actions/runs/28413758338>
- Official trainer: https://github.com/nerfstudio-project/gsplat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/simple_trainer.py
- Official COLMAP parser: <https://github.com/nerfstudio-project/gsplat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/datasets/colmap.py>
- Versioned Prueba02 record: docs/validation-runs/2026-06-29-prueba02-trainer-e2e.md
- Final operator attestation: docs/validation-runs/2026-07-04-trainer-v0.1-final-operator-attestation.md
- Versioned SuperSplat evidence: docs/evidence/prueba02-supersplat.png

19. Freeze declaration

The Trainer v0.1 final baseline is runtime 221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3 under digest sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98. Its real flow is operator-confirmed and its primary execution records were not preserved. Prueba02 original remains separate primary evidence for runtime 9019625 and historical digest sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89.

This is a smoke-validated baseline, not a quality-certified release.